

Method of Handling Traffic — Crew Instructions

Project Information

- **Speed limit:** 45 mph
- **Road type:** Urban (high-speed)
- **Closure type:** Shoulder Closure — Divided Highway
- **Work zone length:** 5,000 ft
- **Jurisdiction:** Englewood
- **Generated:** 2026-08-22

Required Equipment

- 42× Construction/Warning Sign (see sign list below)
 - 2× G20-1 ROAD CONSTRUCTION (NEXT 5000 FT)
 - 2× G20-2 END ROAD WORK
 - 10× G20-5P CONSTRUCTION ZONE plaque
 - 2× R2-10 BEGIN DOUBLE FINES ZONE
 - 2× R2-11 END DOUBLE FINES ZONE
 - 2× R2-1 SPEED LIMIT 25 (work-zone speed posting)
 - 2× R2-1 SPEED LIMIT 45 (posted-speed restoration)
 - 6× R2-6P FINES DOUBLE plaque
 - 2× R9-9 SIDEWALK CLOSED
 - 2× S1-1 SCHOOL
 - 2× W20-1 ROAD WORK AHEAD
 - 2× W20-2 ROAD WORK 700 FT
 - 2× W21-5aR RIGHT SHOULDER CLOSED AHEAD
 - 2× W3-5 ADVISORY SPEED 25
 - 2× W3-5 ADVISORY SPEED 30
- 5× Channelizing Drum (36-inch)
- 59× Traffic Cone (36-inch)
- 5× Type C Steady-Burn Warning Light
- 4× Type III Barricade
- 1× Arrow Board
- 1× Portable Light Plant

Sign Placement Schedule

Sign Code	Description	Distance from Taper
S1-1	SCHOOL	1,550 ft upstream
W20-1	ROAD WORK AHEAD	1,050 ft upstream
W20-2	ROAD WORK 700 FT	700 ft upstream
W21-5aR	RIGHT SHOULDER CLOSED AHEAD	350 ft upstream
G20-5P	CONSTRUCTION ZONE plaque	Within work zone
G20-2	END ROAD WORK	Downstream of work zone
W3-5(30)	ADVISORY SPEED 30	1,060 ft upstream of R2-10
W3-5(25)	ADVISORY SPEED 25	530 ft upstream of R2-10
R2-1	SPEED LIMIT 25 (work-zone speed posting)	Within work zone (paired with G20-5P plaque)
R2-10	BEGIN DOUBLE FINES ZONE	500 ft upstream of work zone
R2-11	END DOUBLE FINES ZONE	500 ft downstream of work zone
G20-5P / R2-6P	WORK ZONE / FINES DOUBLE assembly	2,640 ft assemblies between R2-10 and R2-11
R2-1	SPEED LIMIT 45 (posted-speed restoration)	1,000 ft downstream of work zone

Setup Procedure

Important: Set up from downstream to upstream (against traffic flow). This ensures you are always working behind protection.

1. Position arrow board at the upstream start of the shoulder taper (station 5,510 ft). Set to RIGHT ARROW mode (for right shoulder closure). Activate flashing.
2. Place the END ROAD WORK sign (G20-2) at the downstream termination (station -150 ft), right side of road 46 ft from centerline. Place matching sign on left side.
3. Place 2 traffic cones in the downstream taper, between station -50 ft (shoulder edge) and station 0 (lane edge) at 25 ft spacing — this taper returns traffic from the lane edge back to the shoulder edge past the work zone (MUTCD §6B.08 downstream taper). Then place 57 traffic cones along the right lane edge at 89 ft spacing, starting at station 0 (downstream end of work zone) and working upstream to station 5,000 ft. Offset: 42 ft from centerline.
4. Place G20-5P WORK ZONE plaque(s) within the work zone at approximately 1,000 ft intervals. Place matching plaques on left side.
5. Place 5 channelizing drums in the shoulder taper, starting at station 5,360 ft and working upstream to station 5,510 ft. Space drums at 38 ft intervals. Drums transition from the lane edge (42 ft) to the shoulder edge (52 ft).
6. Place advance warning signs upstream of the taper: Place matching signs on left side at each location.
 - W21-5aR at 350 ft upstream of taper start (station 5,860 ft), right side.
 - W20-2 at 700 ft upstream of taper start (station 6,210 ft), right side.
 - W20-1 at 1,050 ft upstream of taper start (station 6,560 ft), right side.
 - S1-1 at 1,550 ft upstream of taper start (station 7,060 ft), right side.

Takedown Procedure

Important: Remove from upstream to downstream (with traffic flow). Remove advance warning signs first, work zone devices last.

1. Remove advance warning signs: S1-1 first, then W20-1, then W20-2, then W21-5aR. Remove both sides at each station.
2. Remove taper drums from upstream to downstream.
3. Deactivate and remove arrow board.
4. Remove work zone cones from upstream to downstream.
5. Remove G20-5P plaques.
6. Remove G20-2 END ROAD WORK sign(s).

Safety Notes

- All personnel in the work zone must wear ANSI Class 2 (daytime) or Class 3 (nighttime) high-visibility apparel.
- Nighttime operation: Ensure all devices meet retroreflectivity requirements. Flagger stations require 500W quartz illumination at minimum 8 ft height per CDOT S-630-1 Sheet 2 General Note 22.
- Divided highway: All warning and regulatory signs must be posted on BOTH sides of the roadway per CDOT S-630-1 Sheet 2 General Note 8.
- Maintain 360 ft longitudinal buffer space between the taper and the work area at all times.
- Maximum speed reduction per sign installation: 15 mph (CDOT S-630-1 Sheet 2 General Note 3).
- This plan was generated by MHT Tool. A certified Traffic Control Supervisor (TCS) must review, approve, and sign before field implementation.
- Reference: CDOT S-630-1, MUTCD 11th Edition Part 6, Colorado Supplement (effective January 18, 2026).

Pedestrian and Bicycle Accommodation

- A pedestrian facility is affected by this closure. 2 R9-9 SIDEWALK CLOSED signs are placed at the closure (MUTCD §6G.10 — the plain R9-9 legend; no alternate-side assertion, which belongs to R9-10, and no detour-routing signs, which require sidewalk geometry this plan does not carry). Sidewalk closure points are barricaded (MUTCD §6C.02). Pedestrians must not be diverted into an open travel lane.
- **Englewood TC-5B, note 2** — This road draws more than 2 total lanes. Verbatim: "PEDESTRIANS SHALL NOT BE DIRECTED PEDESTRIAN ACROSS A ROAD WITH MORE THAN 2 LANES WITHOUT AN EXISTING MARKED CROSSWALK." (typo in the original sheet). Route any pedestrian crossing via an existing marked crosswalk.
- The TCS verifies pedestrian and bicycle accommodation on site; where a facility is maintained, it must remain accessible and detectable per MUTCD standards.

Emergency Access

- All travel lanes remain open — the closure occupies the shoulder only. Emergency vehicles pass through the work zone in the open lanes.
- The Traffic Control Supervisor completes and confirms the emergency access plan for this site and coordinates notification of police, fire, and emergency medical responder agencies prior to work (TCS duties per CDOT Standard Specifications 630.11).
- An emergency access plan is a required minimum element of the approved Method of Handling Traffic (CDOT 630.10(a)); this section is draft input for it, not the approved plan.

Site-Specific Notes

- **adjacent_intersection** — No devices added — the cross-street approach layout is not generated; see the pending-verification disclosure. (*MUTCD §6N.12 p. 848 — Work within the Traveled Way at an Intersection (11th*

Ed.))

- **adjacent_interchange** — No devices added — the per-ramp interchange layout is not generated; see the pending-verification disclosure. (*MUTCD §6N.16 p. 851 — Interchanges (11th Ed.)*)
- **pedestrian_facility** — Added 4 Type III barricades (sidewalk closure points) and 2 R9-9 SIDEWALK CLOSED signs at the upstream and downstream ends of the work zone. (*MUTCD §6C.02 — pedestrian considerations in work zones*)
- **school_zone** — Added 2 S1-1 SCHOOL signs upstream of the advance warning signs (station 7,060 ft, both sides). (*MUTCD §7B.08 — school zone signing in proximity to work zones*)

Night Operation Notes

- **night_taper_lighting** — Added 5 Type C steady-burn warning lights on taper drums for nighttime visibility. (*MUTCD §6L.07 — warning lights on channelizing devices in nighttime work zones*)
- **night_work_area_lighting** — Added 1 portable light plant at the start of the work zone (25 ft past the buffer) to illuminate the work area. (*CDOT M&S §630.05.1 — illumination requirement for nighttime work zones*)
- **night_retroreflective** — All devices, signs, and barricades must be retroreflective. Verify warehouse inventory before deployment. Inspect retroreflectivity at shift start. (*MUTCD §6F.01 + §6K.01 — retroreflectivity requirements for nighttime visibility*)

Fines Double Operational Rules

CDOT S-630-1 (July 2026) Sheet 12, Fines Double Signing Notes. The Fines Double envelope (W3-5 advisory-speed sign(s) upstream of R2-10, entrance R2-1 posting the reduced limit, R2-10, R2-11, G20-5P/R2-6P assemblies, and the downstream R2-1 posted-speed restoration) ships with this plan because the work-zone posted speed is reduced below the nominal posted speed. Field crew follows the rules below; engineer may adjust placement spacing per Sheet 2 General Note 4.

- **S-630-1 Sheet 12, Note 1** — Install Fines Double signs no more than 4 hours before the start of the work day. Do not leave signs up overnight when work is not active.
- **S-630-1 Sheet 12, Note 2** — Remove or cover Fines Double signs when work concludes; doubled fines apply only when workers or work activity are present in the zone.
- **S-630-1 Sheet 12, Note 3** — Relocate the R2-10/R2-11 envelope to follow the actual work area as the project progresses; do not leave Fines Double signs in place beyond the active work zone.
- **S-630-1 Sheet 12, Note 4** — Maintain a 250 ft minimum spacing between Fines Double signs (R2-10, R2-11, G20-5P/R2-6P assemblies) and other warning or regulatory signs. Engineer may adjust placement to satisfy this minimum.